UNIVERSITÉ D'ALGER 1

Academic Year 2025–2026

A Comparative Study of RAG Techniques for Generating Image Captions

Phase 1: Dataset Exploration and Preprocessing

Dataset, metadata, embeddings, and retrieval infrastructure

Students	Belarbia Tarek; Benamor Mohamed Wacim; Astite Aymen; Wail Bougaada
Institution	Université d'Alger 1
Academic year	2025–2026

Abstract

This report presents Phase 1 of a project on retrieval-augmented image captioning using 10,000 training, 100 validation, and 1,000 test images from MS COCO 2017. The 11,100 selected images have 55,500 reference captions. The pipeline preprocesses images, preserves raw captions, organizes metadata, creates normalized 512-dimensional CLIP embeddings, and builds training-only FAISS indexes. Pipeline quality checks pass for the implemented tests, while overall assignment compliance remains **PARTIAL**. A top-5 retrieval smoke test on 20 held-out queries had zero self-matches and mean top-1 cosine similarity 0.8260; this is not a caption-quality measure. Caption generation, RAG comparisons, and statistical evaluation remain future work.

Contents

Abstract i

1 Introduction and Project Objectives 1

2 Dataset Description and Splitting 1

3 Data Exploration and Statistical Analysis 2

4 Image and Caption Preprocessing 6

5 Metadata Extraction 7

6 Image and Text Embeddings 8

7 Knowledge Base and FAISS Indexing 8

8 Retrieval and Quality Control 8

9 Results and Discussion 10

10 Limitations and Conclusion 11

A Reproducibility Summary 12

1 Introduction and Project Objectives

Image captioning is the task of describing an image in a short natural-language sentence. A useful caption names important objects and may describe their attributes, actions, or relationships. A caption should also avoid details that are not supported by the image. Current vision-language systems can produce fluent text, but fluent wording alone does not guarantee that the details are correct.

Retrieval-Augmented Generation (RAG) is a design in which a system retrieves information from a separate collection and uses it when generating an answer [3]. For image captioning, the retrieved material could include visually similar training images, captions written for those images, or structured visual metadata. The retrieved examples may guide a caption generator, but they may also introduce incorrect objects or relationships. Their effect must be measured rather than assumed.

1.1 Objective

The project asks how retrieval techniques affect caption quality, relevance, and grounding. This report covers Phase 1: the data, metadata, embedding, and retrieval foundation. It does not generate captions or compare RAG methods.

2 Dataset Description and Splitting

2.1 Source and selected splits

MS COCO is a dataset of everyday scenes with object-instance annotations and human-written captions [1, 4]. The project uses the 2017 training and validation sources. The full train2017 annotation audit contains 118,287 images, 860,001 object instances, and 591,753 captions. The val2017 audit contains 5,000 images, 36,781 object instances, and 25,014 captions.

The frozen project subset contains 10,000 train images selected from train2017, and 100 validation plus 1,000 test images selected from val2017. Every selected image has exactly five reference captions. The train split supplies the retrieval knowledge base; validation is available for tuning, and test is held out for evaluation.

Table 1: Frozen project splits and reference-caption counts.

Split	Images	Captions	Refs./image
Train	10,000	50,000	5
Validation	100	500	5
Test	1,000	5,000	5
Total	**11,100**	**55,500**	**5**

The manifests are frozen in JSON and CSV under `data/selected/`. A rarity-first multilabel

selection procedure uses seed 20261106, with deterministic offsets for the train and validation selections. Selection favors coverage of object categories, annotation-count complexity, and object-count groupings; it does not guarantee representativeness.

2.2 References and leakage controls

Each selected image has five references. Caption records preserve raw and cleaned fields; the caption index uses raw training captions only. Both FAISS indexes contain training items. The manifests have no image-ID overlap, and retrieval removes a query's own ID. Test references must remain outside retrieval and generation evidence.

2.3 Test-size note

The original assignment requirements specify 1,000 test images in the detailed dataset and preprocessing instructions. Although one occurrence contains the typographical value `1,00`, the detailed requirements and data-audit instructions establish 1,000 as the intended test-set size. Therefore, this project uses 1,000 test images, consistent with the detailed requirements.

3 Data Exploration and Statistical Analysis

3.1 Object categories and scene complexity

COCO's train2017 annotations cover 80 object categories. The most common annotated class is *person*, with 262,465 instances (30.52% of all 860,001 instances). It is followed by *car* (43,867; 5.10%) and *chair* (38,491; 4.48%). The long-tailed category distribution is shown in figure 1. This analysis describes annotation frequency, not how often objects appear in every image pixel.

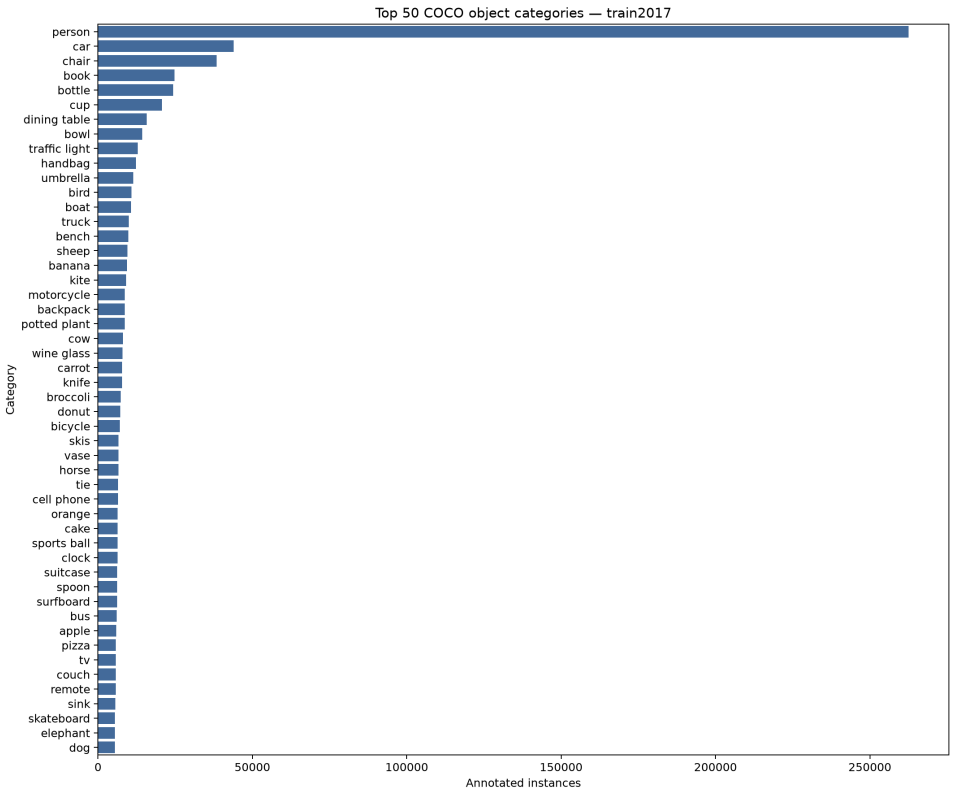


Figure 1: The 50 most frequent object classes in train2017 COCO instance annotations. Counts are annotated object instances.

Across all train2017 images, the mean number of annotated instances is 7.27 and the median is 4; counts range from 0 to 93. The project assigns simple, medium, or complex labels using quartiles of nonzero object counts (Q1 = 2, Q3 = 10). Images with no instances are treated as unknown for this classification. These are annotation-count summaries, not a learned measure of visual difficulty.

3.2 Caption lengths and vocabulary

Across 55,500 raw reference captions, the documented tokenizer counts 591,664 tokens and 9,793 distinct vocabulary items. The tokenizer applies Unicode NFKC normalization and case-folding, retains internal apostrophes, and does not stem or lemmatize. Captions average 10.66 tokens (median 10; range 6–49). Token counts depend on this explicit tokenizer, so results from a different tokenizer may differ.

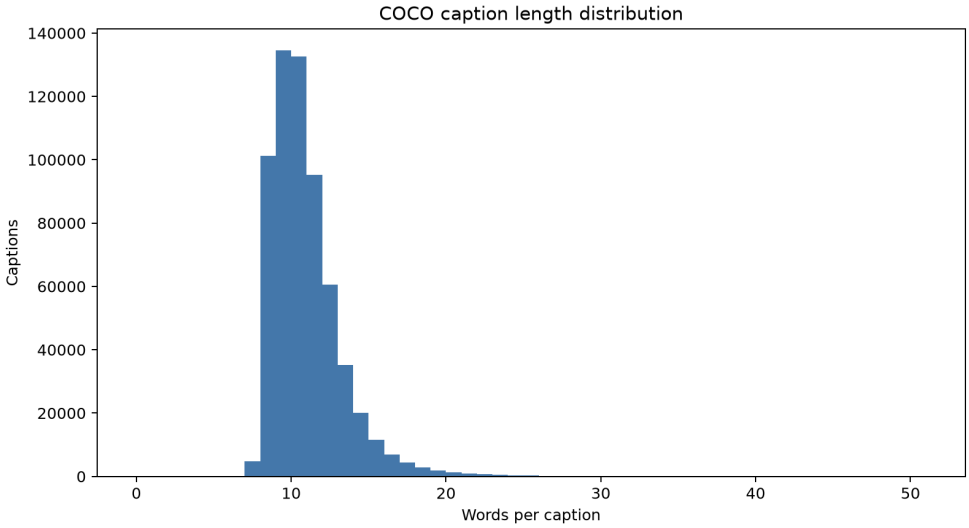


Figure 2: Caption length distribution in the saved analysis. Length is measured in tokens using the project caption-analysis procedure.

The most frequent token is *a* (92,970 occurrences), followed by *of* (19,889) and *on* (18,767). The vocabulary rank-frequency curve illustrates that a small number of words occur very often, while many words are relatively rare.

3.3 Part-of-speech analysis

The project used spaCy with the `en_core_web_sm` model to assign part-of-speech (POS) tags to all 55,500 reference captions [2]. The analysis reports 642,147 model-tokenized tokens with no processing errors. Nouns make up 33.07% of the tagged tokens, determiners 18.54%, and adpositions 15.40%. POS tags are model predictions, not manually checked labels. The counts also use spaCy tokenization, which differs from the tokenizer used for the 591,664-token vocabulary result.

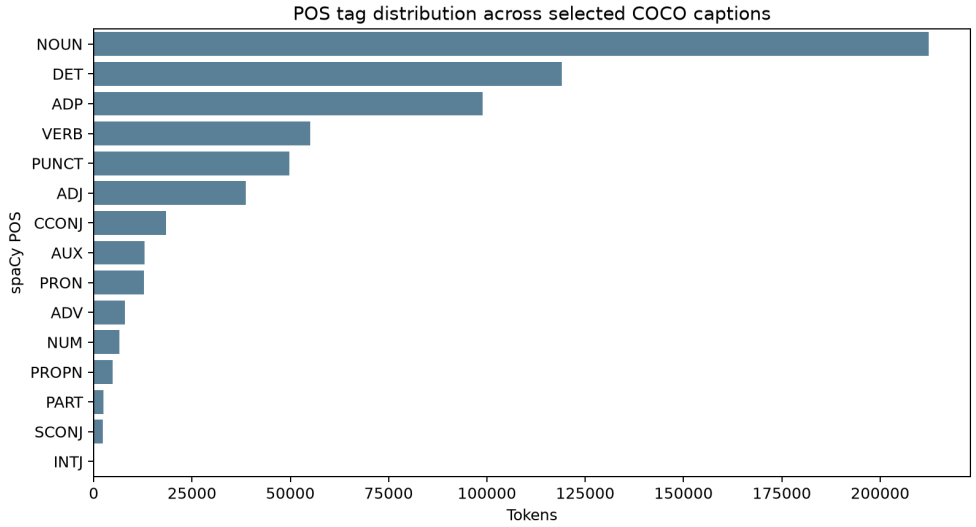


Figure 3: spaCy part-of-speech distribution across selected reference captions. Percentages describe model-tagged tokens.

3.4 Color and image-level visual diversity

The saved color analysis samples pixels and assigns them to coarse color families. Among the 11,100 selected images, red is the most frequent dominant family (4,430 images), followed by gray (3,948) and orange (883). These labels summarize pixels in an image; they do not tell us the color of a particular annotated object. The project also records luminance, saturation, and a global edge-density proxy. No independent, validated shape-recognition analysis is available: edge density reflects contours and texture but does not identify object shapes.

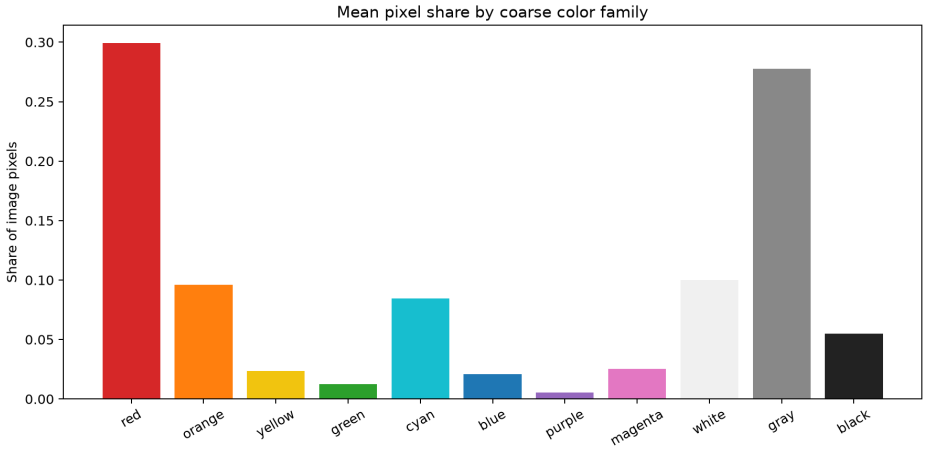


Figure 4: Mean pixel share by coarse color family across the selected images. This is a pixel-level summary, not object-color recognition.

3.5 Scene and spatial summaries

CLIP zero-shot predictions cover indoor/outdoor context, day/night, and scene labels for 11,100 images. The analysis marked 197 context, 582 time, and 1,063 scene predictions uncertain. These are model outputs with relative, uncalibrated scores—not COCO annotations. Box-derived geometric relations include 680,482 left-of, 676,706 right-of, 671,396 below, 584,569 above, 151,587 near, 47,381 overlapping, 28,366 contains, and 30,359 inside pairs. These relations describe box geometry, not actions or semantic interactions.

4 Image and Caption Preprocessing

4.1 Image pipeline

The preprocessing source reads the selected COCO images and applies the following operations:

1. Open and decode each source image.
2. Apply EXIF orientation correction with Pillow's `ImageOps.exif_transpose`.
3. Convert the image to RGB.
4. Resize it with aspect ratio preserved, using Pillow's thumbnail operation and LANCZOS resampling.
5. Center it on a black canvas to produce a 224 by 224 pixel image.

The quality-control report records 11,100 processed files decoded successfully, with no missing files. Each processed image is checked for 224 by 224 dimensions. Encoder normalization is applied when CLIP processes an image; the mean and standard deviation are not baked into the saved JPEG.

4.2 Resize choice

A geometry-only comparison on 100 validation images selected letterboxing: it preserves the full image and aspect ratio. Mean padding was 0.2776; stretching had mean absolute log aspect distortion 0.3309; center cropping discarded a mean 0.2776 of image area. This comparison did not measure model or caption performance.

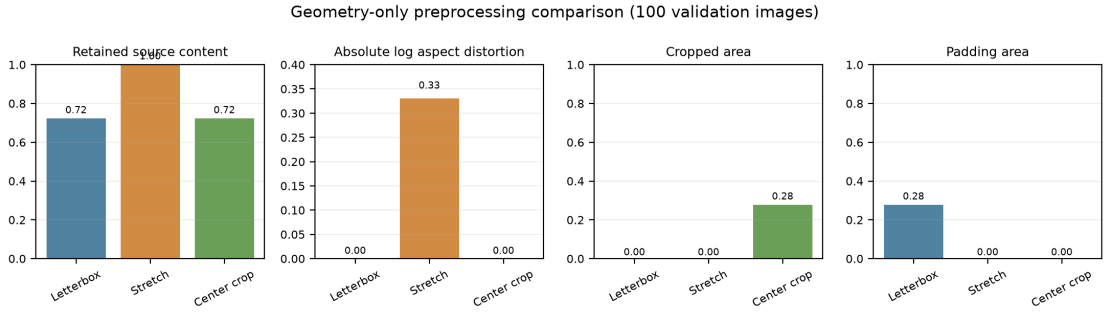


Figure 5: Recorded geometry-only comparison of three 224 by 224 preprocessing configurations on 100 validation images.

4.3 Caption handling

Each selected image has five caption records. `caption_raw` preserves the original reference caption as supplied. The separate `caption_cleaned` field collapses repeated whitespace, trims the ends, and case-folds the text for indexing. The current caption embeddings are generated from raw training captions, not the cleaned field. Future evaluation must use the original raw reference captions.

The source applies EXIF correction and RGB conversion, but historical counts of changed orientations or converted images were not retained. The compliance audit also marks exact source-to-caption mapping verification as partial.

5 Metadata Extraction

5.1 Fields and provenance

Each image record links its COCO ID, filename, split, dimensions, and object records. Other fields come from pixel analysis, box geometry, or CLIP predictions; these sources are not interchangeable. In particular, a CLIP score is not a calibrated probability.

Table 2: Examples of metadata fields and their provenance.

Metadata field	Example	Source
Object category	Person	COCO annotation
Bounding box	`[x, y, width, height]`	COCO annotation
Dominant color	Red	Pixel analysis
Scene context	Indoor	CLIP prediction
Spatial relation	Left of	Bounding-box geometry

6 Image and Text Embeddings

An embedding is a numeric vector representing an image or text for similarity search. The project uses pretrained CLIP ViT-B/32, `openai/clip-vit-base-patch32` [5], to encode images and training captions. Unit-length vectors allow inner-product search to be interpreted as cosine similarity. The model revision, vector settings, environment details, and saved outputs are summarized in table 6.

Image and caption vectors are saved as NumPy `.npy` arrays and mapped to frozen IDs. QC verifies expected counts and dimensions, finite values, unit norms, and row order; it does not establish semantic correctness.

7 Knowledge Base and FAISS Indexing

7.1 Training-only memory

The knowledge base contains CLIP vectors for 10,000 training images and 50,000 training captions, linked to IDs and metadata. Validation and test references are excluded to support leakage-safe experiments.

7.2 Exact search

FAISS stores and searches vectors. Both indexes use `IndexFlatIP`, a flat, exact inner-product index. With unit-normalized vectors, inner product equals cosine similarity:

$$\cos(q, v) = q^\top v \qquad \text{when } \|q\|_2 = \|v\|_2 = 1.$$

The image index contains 10,000 vectors of dimension 512. The caption index contains 50,000 vectors of dimension 512. The index metadata records the source split, model, vector dimension, metric, and mappings. The recorded file sizes are 20,480,045 bytes for the image index and 102,400,045 bytes for the caption index.

Table 3: Training-only vector indexes.

Index	Items	Dimension	Search measure	Source
Image	10,000	512	Inner product = cosine	Train images
Caption	50,000	512	Inner product = cosine	Train captions

7.3 Pipeline overview

Future prompt construction and caption generation will need their own leakage checks.

8 Retrieval and Quality Control

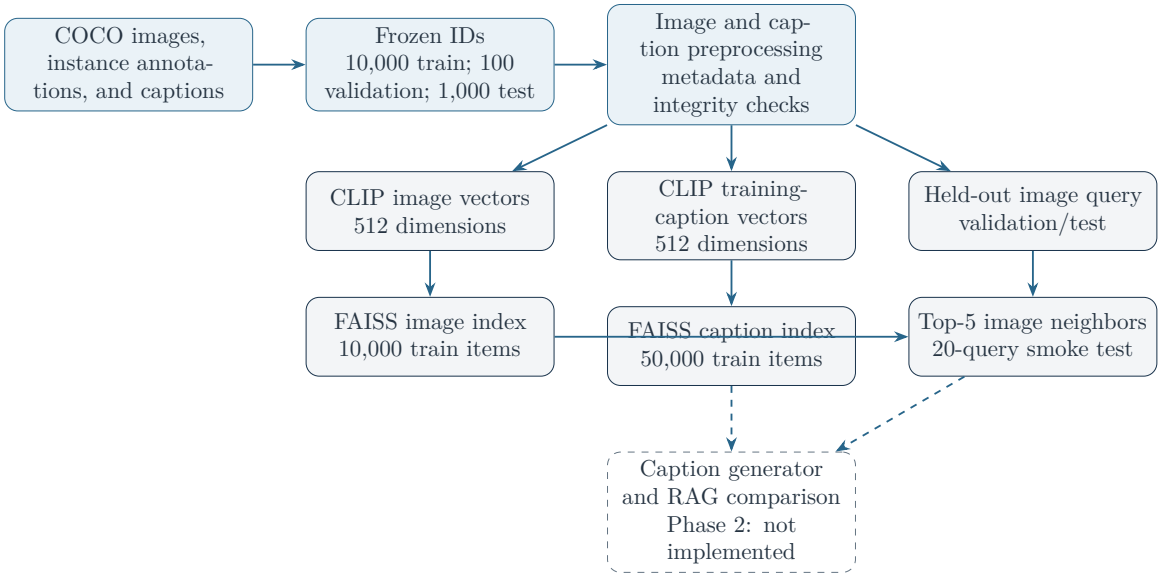


Figure 6: Phase 1 data and retrieval flow. Solid boxes and arrows represent implemented components. The dashed generation component is future work.

8.1 Retrieval smoke test

Each held-out query's saved CLIP image vector searches the 10,000-image training index. The evaluator sampled 20 held-out validation/test images, requested five neighbors, and excluded any same-ID result.

The test returned results for all 20 queries, with zero self-matches and mean top-1 cosine similarity 0.8260. This is a functional check, not a relevance benchmark: without judged neighbors, the similarity is not accuracy, caption quality, or evidence of visual grounding.

Table 4: Recorded held-out image retrieval smoke test.

Measure	Recorded result
Training gallery	10,000 images
Held-out queries	20
Neighbors requested	5 per query
Queries with results	20 / 20
Self-matches	0
Mean top-1 cosine similarity	0.8260

8.2 Quality-control results

The pipeline QC report marks implemented checks PASS. It covers split sizes and separation, five captions per image, image decoding, exact and perceptual duplicate screening, embeddings and mappings, index readback, caption-ID exclusion, and retrieval. Perceptual screening found

nine within-split candidates and zero cross-split candidates at Hamming distance 6; candidates are not proof of duplicates.

Table 5: Selected results from the saved final pipeline quality-control report.

Check	Recorded result
Split sizes	Train 10,000; validation 100; test 1,000.
Reference captions	5 per selected image; 55,500 records total.
Processed image integrity	11,100 decoded; no missing processed images.
Split separation	No image-ID overlap between any pair of splits.
Exact duplicates	0 duplicate groups by SHA-256.
Perceptual-hash screen	9 within-split candidates; 0 cross-split candidates at Hamming distance 6.
Image embeddings	Expected split counts; 512 dimensions; finite and L2-normalized; manifest order checked.
Caption embeddings	50,000 train vectors; 512 dimensions; unique IDs and source order checked.
Caption-index leakage	0 validation/test caption IDs in the training index.
FAISS indexes	Image index: 10,000 vectors; caption index: 50,000; index readback passed.
Image retrieval	20 of 20 queries returned results; top-5; 0 self-matches.

Scope note: The final quality-control report records zero failed, warning, and not-run checks among the checks it implements. The separate requirement-level compliance audit remains PARTIAL.

This PASS applies only to implemented pipeline checks. The separate requirement-level audit rates overall compliance PARTIAL because evidence is incomplete, including unrecorded historical preprocessing counts, partial exact caption mapping verification, and the missing unified environment snapshot. The 1,000-image test size matches the detailed assignment requirements.

9 Results and Discussion

9.1 What the evidence shows

The saved artifacts show that the project assembled the selected COCO splits, processed the images, prepared caption and metadata records, generated CLIP embeddings, built separate training-only indexes, and completed its recorded retrieval and quality-control checks. The category and caption analyses also describe the composition of the selected data. The retrieval test shows that the image index can return neighbors for the sampled held-out queries without returning the query image itself.

9.2 What it does not show

The retrieval similarity mean has no relevance judgments behind it. The category, color, POS, and scene statistics describe the dataset or model outputs; they are not caption-generation outcomes. No generated

captions are available for comparison. Consequently, the current evidence cannot show that one RAG technique is better, that captions are more accurate, or that hallucinations have been reduced.

The train-only memory and frozen held-out splits support a controlled future comparison. Methods should use the same generator and decoding settings, exclude query-associated references, retain per-image outcomes, and pair text metrics with human checks of image support.

10 Limitations and Conclusion

10.1 Limitations and compliance

The requirement-level compliance report marks Phase 1 as **PARTIAL**. The documented limitations are:

- The number of images whose EXIF orientation changed, and the number requiring RGB conversion, were not recorded in historical logs.
- Caption records preserve raw and cleaned fields, but the compliance audit marks exact source-to-caption mapping verification as partial.
- A unified run record with the complete Python environment and resolved dependency versions is missing.

COCO object annotations are not exhaustive descriptions of every visible object. Box geometry does not provide semantic relations or actions. CLIP prompt outputs and spaCy POS tags can be wrong, and CLIP scores are not calibrated probabilities. Pixel color and edge density are coarse global summaries; edge density does not recognize object shapes. The selected data are a subset of COCO, and the 20-query retrieval test is a smoke test rather than a broad benchmark.

The quality-control report records `overall_status: PASS` and `phase1_complete: true` for its implemented checks. The separate compliance audit records `overall_status: PARTIAL` against the assignment requirements because of the evidence gaps listed above. These statuses have different scopes; pipeline QC PASS does not establish full assignment compliance.

10.2 Conclusion and future work

Phase 1 established a processed dataset, metadata, normalized CLIP embeddings, train-only FAISS indexes, and basic retrieval checks. Phase 2 remains future work: generate captions, compare retrieval strategies under shared settings, evaluate caption quality and hallucinations, and conduct paired statistical tests and ablations. No caption-generation results or caption metrics are reported here.

A Reproducibility Summary

Table 6: Compact configuration and reproducibility record.

Setting	Recorded value
Selection seed	20261106
Image input	224 by 224 pixels; RGB; black letterbox padding
Image format	JPEG
Image and text model	`openai/clip-vit-base-patch32`
Recorded model revision	`3d74acf9a28c67741b2f4f2ea7635f0aaf6f0268`
Embedding dimension and type	512; float32
Normalization	L2; unit-length vectors
Batch size	64
Text maximum sequence length	77 tokens
Image index	FAISS `IndexFlatIP`; train gallery
Caption index	FAISS `IndexFlatIP`; train captions only
Retrieval settings	top-$k = 5$; 20 held-out query smoke test
POS model	spaCy `en_core_web_sm`
Run environment	CPU; PyTorch 2.15.0+cpu; Transformers 5.22.1
Execution time	1,121.3 seconds for training-caption embeddings
Split manifests	Frozen JSON/CSV files under `data/selected/`
Report build	Tectonic 0.17.0; `tectonic main.tex`

The authoritative parameters are in `configs/config.json`; frozen split IDs are in `data/selected/`. Embeddings, mappings, and FAISS indexes are saved under `embeddings/` and `vector_database/`. The saved caption-vector run records CPU execution, PyTorch 2.15.0+cpu, Transformers 5.22.1, and 1,121.3 seconds. A complete unified environment snapshot was not available, so these versions should not be treated as a full environment lock.

The report was built with Tectonic 0.17.0. From this directory, compile using `tectonic main.tex`; the Python pipeline itself is not rerun to build the report.

References

[1] COCO Consortium. Coco dataset and annotation documentation. Dataset documentation, 2026. URL https://cocodataset.org/. Accessed November 2026.

[2] Explosion AI. spacy: Industrial-strength natural language processing in python. Software documentation, 2026. URL https://spacy.io/usage. Accessed November 2026.

[3] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474, 2020. URL https://arxiv.org/abs/2005.11401.

[4] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. Microsoft coco: Common objects in context. In *European Conference on Computer Vision*, pages 740–755. Springer, 2014. doi: 10.1007/978-3-319-10602-1_48.

[5] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 8748–8763, 2021. URL https://proceedings.mlr.press/v139/radford21a.html.